A Mathematical Analysis of the Problem of Question Choice for Rote Memorization

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1 Introduction

The Library is the structure which encodes a hierarchical directory of questions, which have a large number of properties and data associated with them, and which are meant to be memorized by the user via repetition.

The APIs which utilize this data structure to ask the end-user questions and allow them to be answered are called the uncommon state managers because they use separate internal state models, but none of this state is present in the common transmissible format of the Library structure itself. They implement the `Askable` interface and can therefore also be called "askables".

The critical problem is what question to show the user next. The simple strategies are as such: The `TestLibrary` askable shuffles all questions and presents them to the user in that order. When the list is exhausted, the process begins again. The `UniformLibrary` simply selects a question at random, with all questions having equal weight. These strategies are simple and often effective, but fail on large sets and may be overwhelming for users with next to no familiarity with the material. The `AdaptiveLibrary` uses algorithms based on the math in this document to choose questions in such a way **that the perceived difficulty is held constant over time**.

2 Difficulty Approximation

A naive approach to approximating the user's difficulty with a question is to assign a value of 1 to their correct answers and 0 to incorrect answers. Then, we average their answers to get the fraction of the time that they answered correct. We can infer that this value approximately gives the percent chance that they will answer correctly the next time they are asked. The problem is that we are ignoring that they are learning over time and therefore the later answers should be given more weight. A more elegant and accurate solution is to use a moving average, where their latest answer is averaged using a constant weight with their previous average - henceforth the mastery level. The weight of the new answer ranges from 0 to 1 and is called the adaptation rate and denoted A_r. Therefore, the new mastery level M_n is calculated as such:

$$M_n = M_{n-1} \cdot (1 - A_r) + A_{n-1} \cdot A_r$$

where A is the sequence of answers - a sequence of values in the range $[0, 1]$ - and M is of course the sequence of mastery values beginning with some initial mastery value M_0. This gives exponentially decreasing weight to successively older samples. For example if $A_r = 0.5$, then the most recent answer has a weight of 0.5, whereas the previous has a weight of 0.25 and the one before that 0.125 and so on. The sequence M therefore serves as an approximation of the percent chance that the next answer will be correct.

By taking an average of all the active questions' mastery levels, weighted by their probability of being chosen, we obtain the probability that the next question will be answered correctly before we know what that question is. The complement of this value can be called the *difficulty*. **A difficulty near 1 means that the user is getting many questions wrong. They are likely overwhelmed and frustrated. A difficulty near 0 means that they are getting everything correct. They are probably bored and may not be learning anything at all**.

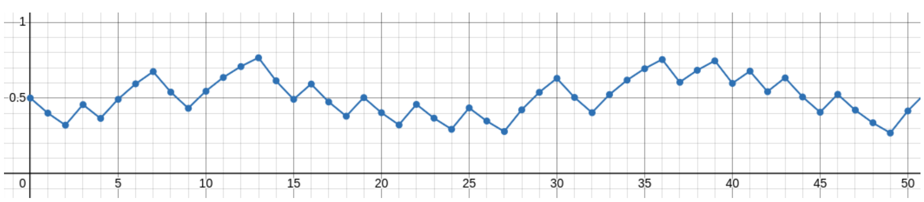

Figure 1: A sequence of 50 mastery levels generated from an answer sequence whose samples are each 1 or 0 with equal probability. Note how consecutive correct or incorrect answers yield diminishing returns.

2.1 Initial Mastery

There is a strong question as to what should be the initial mastery. The truth is that this system fails to incorporate any kind of error-approximation. In order to effectively introduce users to the system, instead of assuming their mastery level, the weight of the question is "pulled-high" as discussed in the next section, so that the initial mastery is irrelevant.

2.2 Correct Guesses

Multiple-choice and particularly true-false question can be correctly guessed a significant fraction of the time. Therefore, we adjust the adaptation rate downwards to compensate and therefore converge more slowly for these kinds of questions. The adaptation is multiplied by the complement of the probability of a correct guess. So, for a multiple-choice question with four choices the true A_r is $1 - 0.25 = 0.75$ times, or three fourths of the base A_r. For true-false questions it is one half the base A_r.

2.3 Partial Credit

3 Question Weight

A certain degree of randomness or "jitter" is necessary in the system. By weighting the questions differently, we nudge the distribution of the asked questions so as to encourage learning while discouraging both frustration and boredom. We would like to ensure that long-mastered questions are still asked every so often while ensuring the bulk of the questions are new to the user.

I imagine the weights of the questions - when sorted - form a geometric series, just like the weights of the answers towards the mastery value. However, this can't be an enforced property of the weights because it is unduly computationally complex (we should not need to sort anything) and would also assign different weights to questions with equal mastery level. Such unfavorable properties are a sure sign that we're straying from the correct path.

In order for the geometric relationship between weights to be feasible, the range of valid weights must extend infinitely.

We can imagine the range of mastery levels divided into strata. The boundaries separating these regions correspond to the mastery levels which can be obtained by answer sequences consisting either of an arbitrary number of correct answers or an arbitrary number of incorrect answers. They are sparse in the middle and approach infinite density at either edge.

3.1 Repeat Questions

It is not preferable to ask someone a question twice in a row. As such, the base object (from which the uncommon state managers are derived) implements a prevention mechanism with two prongs. The first is that it avoids selecting the previously chosen question, and the second is that it forcefully windows a question if the window only contains a single question, and that question was asked last. Only if there is a single selected question will it allow a question to appear twice in a row.

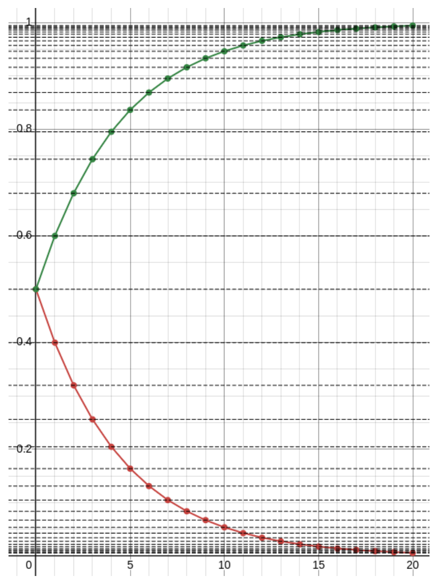

Figure 2: Two sequences of mastery levels - a green one formed by an answer sequence of all 1s, and a red one formed by an answer sequence of all 0s. Note the strata delineated by the black lines which pass through all the values in either sequence of mastery values.